

Executive Work Experience & Strategic IT Capability Presentation

Candidate: Maheshchandra Hegde | Position: Senior Manager – IT

A comprehensive portfolio mapping enterprise technology leadership, mission-critical infrastructure, cybersecurity governance, and operational resilience to Obayashi's global engineering mandates.

IT Operations & Infra

High availability, hybrid cloud, disaster recovery & multi-site continuity

Cybersecurity & Risk

Zero Trust, access controls, vulnerability mitigation & ISO compliance

Project & Team Delivery

Enterprise systems, cross-functional squads, digital transformation

Vendor & Budget Control

SLA governance, contract management, resource optimization & CapEx/OpEx

Strategic Leadership Profile & Alignment with Obayashi Corporation

Maheshchandra Hegde

Technology Leader · Systems Architect · Enterprise Delivery Expert

- **Total Experience:**
20+ Years leading enterprise IT, architectures & engineering teams
- **Education:**
MS in Computing (Robert Gordon Univ., UK) · B.E. in Electronics & Communication
- **Key Past Tenures:**
Cyient (Philips Critical Care), Technoyana, UST Global (Cisco Systems), Moonraft, Ness, ThoughtFocus, Cognizant
- **Core Competencies:**
Hybrid Infrastructure, IT Operations, Mission-Critical Availability, Cybersecurity, Team Mentorship, Vendor SLAs
- **Industry Domain:**
Healthcare Critical Systems, Enterprise Networking & Dashboards, High-Reliability SaaS, 3D/Spatial Tech

1. Multi-Site Infrastructure & High Availability

Proven expertise running mission-critical platforms demanding 99.95%+ uptime. Direct experience engineering resilient edge systems that guarantee business continuity even during remote network dropouts.

2. Cybersecurity Governance & Risk Mitigation

Strict adherence to compliance frameworks, role-based access governance (RBAC), data protection policies, secure CI/CD pipelines, and active threat vulnerability management.

3. Digital Transformation & Enterprise Systems

Spearheaded enterprise system integrations, API gateways, automated telemetry, and modernized legacy workflows into performant, scalable cloud architectures.

4. Team Empowerment, Vendor Governance & Fiscal Discipline

Led cross-functional squads of engineers, QA, and analysts. Successfully managed third-party vendor contracts, cloud budgets (AWS/Azure/GCP), and SLA compliance.

Project Overview & Objectives: Mission-Critical Enterprise System (Philips/Cyient)

Project Profile: Philips ICCA

Intellispace Critical Care & Anesthesia Platform

Role:

Senior Technical Lead / Systems Architect

Tenure:

Nov 2023 - Mar 2025

Scale:

Enterprise Global Hospital Deployments & High-Availability Clinical ICU Units

Domain Context:

Life-critical, real-time patient monitoring, zero-latency clinical alerts & strict regulatory compliance

Tech Ecosystem:

React, Node.js, Micro-frontends, REST/WebSocket APIs, Secure Auth, Enterprise Telemetry

Strategic Objective 1: High Availability & Zero-Fault Operational Resilience

Architect and deliver web-tier systems for ICU and anesthesia workflows where system downtime is life-threatening. Established fail-safe fallback rendering, continuous health-check telemetry, and zero unhandled state exceptions across 24/7 continuous operations.

Strategic Objective 2: Data Protection, Security & Regulatory Compliance

Enforce strict healthcare information security standards (HIPAA, FDA software safety benchmarks, ISO 27001 compliance). Enforced end-to-end data encryption in transit and at rest, granular role-based access (RBAC), and full audit logging of administrative and user actions.

Strategic Objective 3: Scalable Modular Architecture & Seamless Integration

Transitioned monolithic legacy interfaces into a decoupled, micro-frontend architecture with standardized APIs. Enabled independent modular deployments, drastically reduced release risk, and unified clinical data feeds from diverse hardware monitors.

Project Overview: High-Availability Cloud Transformation & Edge Resilience (Technoyana)

Technoyana Platforms

Digital Transformation & Distributed SaaS Ecosystem

Role:

Co-Founder & Director of Technology / Architecture Lead

Tenure:

May 2021 - Oct 2023 (Ongoing Strategic Advisor)

Key Platforms:

Twitan Sports OS (Shutlify), Fintech Operations Engine, Spatial AI Pipelines (Srushtilabs)

Operational Reality:

High-concurrency events, remote arenas/venues with degraded cellular connectivity, multi-tenant cloud infra

Relevance to Obayashi:

Directly mirrors remote construction site challenges: volatile network connectivity, field staff data capture, automated cloud sync

1. Edge-Resilient 'Offline-First' Architecture for Unstable Networks

Architected an offline-first data model utilizing local storage, background Service Workers, and optimistic transactional queuing. When on-site network drops occur, field users continue full operations uninterrupted; bi-directional conflict resolution synchronizes cloud data the moment connectivity is restored.

2. Cloud Infrastructure & Cost Governance (AWS / Firebase / Serverless)

Directed cloud infrastructure roadmap, automated CI/CD pipelines, containerized deployments, and disaster recovery replication. Implemented automated resource scaling and log archiving, cutting monthly cloud spend by 22% while handling 10x traffic bursts.

3. Digital Innovation: Spatial 3D & Automated Asset Workflows

Pioneered spatial technology workflows (Voxelforge / Srushtilabs) producing modular 3D assets, automated mesh optimization, and WebGL visualizations. Establishes deep alignment with Obayashi's BIM, Digital Twin, and 3D construction modeling initiatives.

Your Role & Core Responsibilities: IT Strategy, Operations & Governance

1. IT Strategy, Policy & Enterprise Architecture

- Formulated multi-year technology roadmaps aligning IT investments with organizational growth objectives.
- Established standardized IT policies, coding standards, system architectures, and documentation protocols.
- Conducted architectural governance reviews to ensure maintainability, scalability, and technical debt reduction.
- Evaluated emerging enterprise technologies (cloud, automation, spatial 3D) to drive continuous capability evolution.

2. Infrastructure Operations, Cloud & Business Continuity

- Ensured 24/7 availability and resilience of mission-critical systems and multi-tier network topologies.
- Supervised automated backup regimens, disaster recovery (DR) simulations, and zero-loss business continuity plans.
- Managed hybrid cloud workloads, virtualization environments, network security boundaries, and telemetry.
- Instituted synthetic monitoring and automated alerting to resolve system anomalies before end-user impact.

3. Cybersecurity Controls, Access Management & Risk Mitigation

- Enforced Zero-Trust principles, role-based access control (RBAC), multi-factor authentication (MFA), and audit logging.
- Conducted proactive vulnerability assessments, dependency scanning, code audits, and third-party risk evaluations.
- Partnered with enterprise compliance teams to ensure strict adherence to ISO 27001 and industry safety standards.
- Formulated incident response playbooks, disaster containment procedures, and cybersecurity staff training.

4. Team Leadership, Vendor Governance & Fiscal Management

- Led, mentored, and fostered technical excellence across cross-functional engineering, infrastructure, and QA squads.
- Governed multi-vendor contracts, service provider RFPs, cloud infrastructure budgets, and strict SLA compliance.
- Optimized CapEx and OpEx allocations through license rationalization, automation, and cloud cost management.
- Instituted transparent KPI dashboards tracking uptime, incident resolution MTTR, and engineering delivery velocity.

Major Achievements & Quantifiable IT Milestones

99.95%+

Critical System Availability

Sustained across 24/7 clinical ICU operations and high-traffic event workloads

100%

Data Persistence at Edge

Zero data loss during field network disconnections via offline-first syncing engine

35%

Faster Release Cycles

Achieved by implementing automated CI/CD pipelines, containerization & lint tests

22%

Cloud Cost Optimization

Saved through resource rightsizing, auto-scaling policies, and license rationalization

★ **Enterprise Architecture Modernization & Zero-Downtime Migration**

Successfully transformed legacy monolithic systems into modular, API-driven architectures at Cyient/Philips and Technoyana. Reduced regression cycles from weeks to hours and eliminated deployment downtime, ensuring hospital and enterprise operations remained completely uninterrupted during version upgrades.

★ **Multi-Site Network Resilience & Field Data Integrity Engine**

Designed and deployed an edge-resilient synchronization pipeline for environments plagued by unstable cellular and local networks. Handled thousands of concurrent transactional updates offline with automatic background sync and deterministic conflict resolution upon network restoration.

★ **Enterprise Operational Governance & Multi-Vendor SLA Excellence**

Managed cross-functional technical teams (15+ engineers, QA, DevOps) and vendor relationships with infrastructure providers, software vendors, and hardware suppliers. Standardized SLA benchmarks, improved incident MTTR by 40%, and received multiple corporate awards for excellence.

Challenges Faced & Strategic Resolutions Implemented

Challenge 1: Severe Network Fluctuation & Remote Connectivity Drops

Operating conditions at remote event venues and distributed client sites suffered frequent Wi-Fi packet drops, ISP latency spikes, and dead zones, causing cloud disconnects and user data loss.

Resolution Implemented:

Architected an 'Offline-First' state engine using IndexedDB client persistence and service workers. Implemented optimistic UI updates and a transactional event replay queue that automatically resynchronizes once the network reconnects.

Result: Impact: 100% data preservation, zero operator workflow interruptions, seamless user experience.

Challenge 2: Architectural Fragility & High Downtime Risk in Monolithic Legacy Systems

A legacy codebase had tight coupling between presentation and backend data layers. Deploying bug fixes or new features carried severe risks of regressions and system outages in critical environments.

Resolution Implemented:

Decomposed the monolith into isolated micro-frontends and standardized REST/WebSocket contracts. Established automated unit/integration testing pipelines and canary deployments to validate changes in staging.

Result: Impact: Reduced regression risk to near-zero, cut deployment cycles by 35%, and enabled independent modular upgrades.

Challenge 3: Complex Multi-Vendor Dependencies & SLA Enforcement Gaps

Dispersed third-party cloud tools, software licenses, and hardware vendors lacked centralized SLA tracking, resulting in finger-pointing during service degradation and creeping monthly cloud expenditure.

Resolution Implemented:

Instituted a centralized IT Governance & SLA Dashboard with automated latency/uptime telemetry. Standardized vendor contract SLAs with penalty clauses, monthly executive reviews, and automated idle resource termination.

Result: Impact: Slashed mean-time-to-resolution (MTTR) by 40% and trimmed overall recurring cloud/vendor spend by 22%.

Lessons Learned & Enterprise Best Practices

1. Resilience Must Be Engineered into the Architecture, Not Added as an Afterthought

In critical operations (whether hospitals or construction job sites), network connectivity and hardware will inevitably experience degradation. Systems must be designed with graceful degradation, local caching, and deterministic reconciliation rather than assuming continuous, low-latency broadband.

2. Cybersecurity is an Organizational Culture, Not Merely a Perimeter Firewall

A majority of enterprise security breaches stem from misconfigured access rights, third-party vendor credentials, or unpatched endpoints. Implementing Zero Trust Network Access (ZTNA), strict least-privilege RBAC, and continuous security hygiene reviews protects the organization far more effectively than traditional boundary defenses alone.

3. Proactive Telemetry & Observability Prevents Operational Crises

Waiting for users to report an outage is a failed IT strategy. Instituting synthetic transactions, automated latency monitoring, CPU/memory threshold alerts, and automated runbooks allows IT operations to resolve 80% of potential incidents before end users or project sites notice any disruption.

4. Technical Leadership Requires Bridging Field Realities with Executive Strategy

Successful IT leadership is not about adopting technology for its own sake; it is about solving concrete business pain points. Spending time understanding how field teams, site managers, and business stakeholders operate ensures IT investments deliver measurable productivity, safety, and financial returns.

Suggestions for Future Process Improvements at Obayashi Corporation

1. Standardized Construction Site 'IT-in-a-Box' Rapid Deployment Kit

Opportunity: Setting up reliable IT infrastructure at new construction sites often takes weeks and suffers from inconsistent hardware and security standards.

Proposed Solution: Design a containerized 'Site IT Kit' combining dual-SIM 5G/satellite SD-WAN routers, local edge server/caching node, pre-configured biometric access control, and ruggedized Wi-Fi 6 access points. Enables full site IT commissioning within 48 hours with automated zero-touch provisioning.

2. Cloud BIM & Common Data Environment (CDE) Acceleration Node

Opportunity: Field teams struggle downloading and rendering multi-gigabyte 3D BIM models, CAD drawings, and point clouds over remote job-site cellular connections.

Proposed Solution: Deploy localized edge-caching appliances paired with WebGL-based progressive 3D rendering (leveraging spatial engineering patterns). Site engineers view and markup complex 3D digital twins instantly without saturating job-site uplink bandwidth.

3. Subcontractor Zero-Trust Security Mesh & Automated Vendor Compliance

Opportunity: General contractors interact with hundreds of specialized subcontractors and vendors, introducing major cybersecurity and intellectual property leakage risks.

Proposed Solution: Establish a cloud-native Zero-Trust Network Access (ZTNA) portal. Subcontractors access only designated project folders with time-bound credentials, multi-factor authentication, watermarked document viewers, and automated endpoint compliance checks.

4. AIOps for Predictive IT Maintenance & Automated Disaster Recovery (DR)

Opportunity: Unplanned server outages, ISP disconnections, or backup failures at remote regional hubs disrupt critical construction scheduling and bid deadlines.

Proposed Solution: Implement AI-assisted log analysis and telemetry to detect early indicators of disk failure, bandwidth saturation, or cyber anomalies. Pair with automated immutable cloud backups (ransomware-proof) and scheduled push-button disaster recovery failover drills.

First 90 Days: Strategic Execution Roadmap for Senior Manager – IT

Days 1 – 30: Discovery & Baseline Assessment

Objective: Understand Current State, Build Relationships & Audit Risks

- ✓ Audit all existing IT infrastructure, networks, cloud tenancies, and active construction site connections.
- ✓ Interview business unit heads, site project managers, and executive leadership to identify operational bottlenecks.
- ✓ Review existing cybersecurity controls, backup regimes, DR documentation, and compliance standings.
- ✓ Analyze IT budget utilization, vendor contracts, software licenses, and current SLA performance metrics.

Days 31 – 60: Stabilization & Quick Wins

Objective: Fortify Security, Address Gaps & Streamline Operations

- ✓ Remediate critical vulnerabilities identified during audit; enforce uniform MFA and access policies.
- ✓ Optimize job-site connectivity baselines; resolve recurring network instability pain points at active sites.
- ✓ Consolidate vendor SLAs, eliminate redundant licenses, and introduce transparent IT support ticketing KPIs.
- ✓ Conduct a full table-top Disaster Recovery (DR) and backup verification exercise to guarantee system resilience.

Days 61 – 90: Modernization & Strategic Roadmap

Objective: Drive Scalability, Digital Transformation & Long-Term Value

- ✓ Present a 2-3 Year IT Strategy & Digital Roadmap aligned directly with Obayashi's corporate objectives.
- ✓ Pilot the 'Site IT-in-a-Box' rapid deployment kit for upcoming construction projects.
- ✓ Initiate automation initiatives (CI/CD, automated log telemetry, self-service IT service catalog).
- ✓ Foster continuous team upskilling, mentorship, and technical excellence within the internal IT workforce.

OBUYASHI CORPORATION · IT LEADERSHIP COMMITMENT

Delivering Reliable, Secure & Future-Ready Technology Services

- Strategic Value Creation: Aligning IT roadmaps directly with Obayashi's engineering excellence and corporate vision.
- Operational Dependability: Ensuring 24/7 resilience, zero data loss, and rapid incident resolution across all offices and project job-sites.
- Robust Cybersecurity & Risk Mitigation: Protecting critical blueprints, intellectual property, and supply chain integrity with Zero Trust governance.
- Collaborative Culture & Team Growth: Mentoring high-performing IT talent while upholding Japanese corporate standards of precision, quality, and mutual respect.

Thank you for the opportunity. I look forward to our Final Interview discussion.

Maheshchandra Hegde | hid.mahesh@gmail.com | +91 9535253329 / +91 7022407280